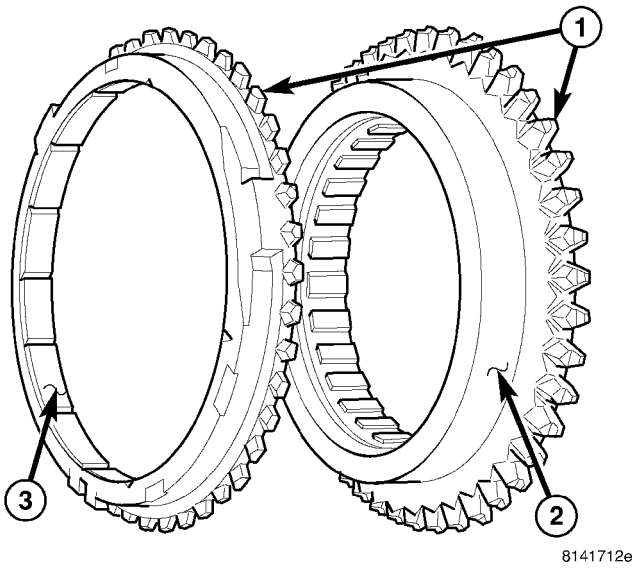

Fig. 55 SYNCHRONIZER RINGS

1 - TEETH
2 - FRICTION SURFACE
3 - FRICTION MATERIAL

Synchronizer components: Inspect for worn, chipped or cracked teeth (1) and burned friction surface (2) or flaking off friction material (3) (Fig. 55). **Synchronizers are serviced as an assembly. 3-4 synchronizer assembly (2) is serviced with countershaft.**

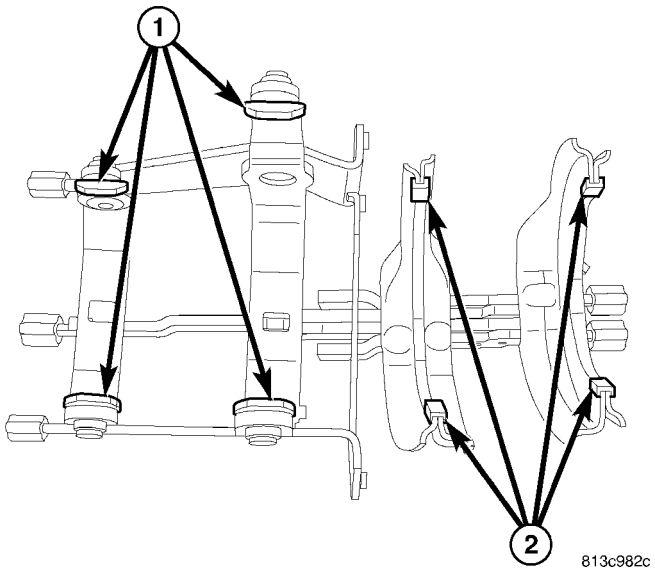

Fig. 56 SHIFT FORKS

1 - SHIFT FORK SHOES
2 - SHIFT FORK PADS

Shift forks: Inspect shift forks and shoes (1) (Fig. 56) for wear and distortion. Check fit of fork shoes in synchronizer sleeve to ensure parts fit and work smoothly, replace if necessary. If shift fork pads (2) are worn, the shift fork must be replaced.

Housing/Tail housing: Inspect sealing and mating surfaces are free of burrs and nicks. Inspect alignment dowels are tight and in good condition.

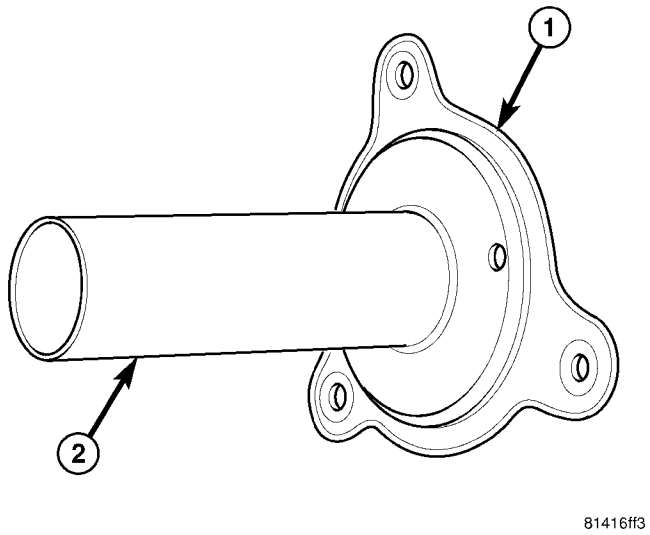

Fig. 57 RETAINER INPUT SHAFT

1 - RETAINER
2 - SLIDE SURFACE

Input Shaft Retainer: Inspect retainer (1) release bearing slide surface (2) (Fig. 57).

ASSEMBLY

WARNING: Use welding gloves when handling heated components. Failure to follow these instructions will result in personal injury.

CAUTION: A bearing heater is used to assembly some components. Use only a bearing heater/hot plate and follow manufactures instructions. Heat components to 100 - 177 Celsius (212° Min - 350° Max Fahrenheit). Never use an open flame to heat components. Never leave components on heater for and extended amount of time. If component is discolored after heating the component has been overheated and must not be used. Failure to follow these instructions will result in component damage.

MAINSHAFT

NOTE: Reuse snap rings in there original locations. Or thickest snap ring that will fit.

(1) Place mainshaft in Fixture 9648 mounted in a vise.